# SIMATS ENGINEERING

## ASSESSMENT TOOL 2 : Scenario-Based

**COURSE NAME:** Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

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**COURSE OUTCOME COVERED**

**CO2:** *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

**Total Marks: 25**

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**Code of Conduct**

(192525271)
I G. Jaahnavi (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: G. Jaahnavi

**Assessment Tasks**
**Scenario-Based Questions**

1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.

2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy. Design a solution using cloud-native backup, replication, and versioning.

Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved.
Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.

5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

**Scenario-Based Rubric**

| Criteria | Weightage | Level 4 - Exemplary | Level 3 - Proficient | Level 2 - Developing | Level 1 - Beginning |
|---|---|---|---|---|---|
| Understanding of Scenario | 20% | Demonstrates complete understanding of the scenario context | Good understanding with minor gaps | Partial understanding | Fails to understand the scenario |
| Identification of Risks / Issues | 20% | Identifies all major issues correctly | Identifies most issues | Limited issue identification | Misses major issues |
| Application of Concepts | 25% | Applies virtualization concepts effectively | Applies concepts with minor mistakes | Partially correct application | Weak application |
| Decision Justification | 20% | Decisions are well justified and technically strong | Reasonable justification | Weak justification | No useful justification |
| Clarity | 15% | Clear and well-organized response | Generally clear | Somewhat unclear | Poorly presented |

CO2- AT2

Scenario - based

Name :- G·Jaahnavi (192525277)

(1) Media company - video streaming platform

Scenario:- A media company wants to migrate its video streaming platform to the cloud. The System must handle millions of concurrent users with minimal buffering. Design a cloud architecture using Scalability & Caching techniques.

Cloud Architecture Design

The proposed cloud architecture consists of the following components:

* Users : Access the streaming platform from the web.
* DNS Service : Routes users to the nearest available cloud region.
* Content Delivery Network (CDN) :- Stores cached copies of popular videos at edge Servers close to users, reducing latency & buffering.
* Load Balancer : Distributes incoming traffic evenly across multiple application servers.
* Auto-scaling group : Automatically adds or removes application Servers based on CPU usage, memory utilization or network traffic

* Application Servers :- process user requests, authentication & video streaming.

* Cache : stores frequently accessed data such as user sessions & video metadata to reduce load.

* cloud object storage :- Stores video files with high durability & availability.

Role of Load Balancing

Load balancing improves performance by

* Distributing user requests evenly among multiple userservers.
* Preventing server overload
* Improving system reliability & availability.

Auto Scaling :- It is automatically increase servers during peak traffic & decreases them during low traffic, ensuring smooth streaming.

Cost optimization :-

* use auto scaling to avoid paying to unusal servers.
* use CDN to reduce bandwidth costs.
* Store old videos in low cost storage.
* use Reserved for lower cloud costs.
* Monitor resource usage & remove idle resources.

conclusion :- A cloud architecture with CDN, load balancing, auto scaling & caching ensures smooth video streaming for millions of users with buffering.

② FinTech Start up - Multi-cloud container Depolyment

Introduction:. A fintech startup is depolying applications using Containers across multiple cloud providers. To ensure Consistent depolyments, scalability the company should use Container orchestration with a multi cloud architecture.

Solution using container orchestration & Multi cloud

* Package applications using Docker containers
* use Kubernetes to manage, depoly & monitor containers.
* Deploy applications across multiple cloud provider (AWS, Azure, GCP) to improve reliabity
* use Infrastructure as code (Iac) tools like Terraform for Conssitent depolyment.

Service Discovery & Scaling :-

* Kubernetes Service Discovery automatically enables communication b/w micro services.
* Horizontal Pod Autoscaler (HPA) automatically increases or decreases containers based on CPU or memory usage.

* Load balancers distribute traffic evenly among containers.

* Self healing automatically resarts failed containers.

**Benefits of Multi cloud**

⇒ High availability & fault tolerance.

⇒ Disaster recovery

⇒ Avoids vendor lock in

⇒ Better Performance by choosing the nearest cloud region.

⇒ Flexibility to use services from different cloud providers.

**Risks of Multi-cloud**

⇒ More complex management

⇒ Higher networking costs

⇒ Security policy management becomes difficulty.

⇒ Data Synchornization challanges.

⇒ Requires skilled administrators.

**Conclusion**:- using Kubernetes & a multi cloud strategy providers consistent depolyments, automatic scaling & high availability.

③ E-Learning Platform - Backup & Disaster Recovery

Introduction:- An e-learning platform lost data due to accidental deletion. A cloud backup & disaster recovery solution is needed to protect data & ensure service continuity.

Cloud-Native Backup, Replication & versioning

* Enable automatic cloud backups at regular interval
* Store backup copies in multiple cloud regions for disaster recovery.
* use data replication to keep a copy of data in another location.
* Enable object versioning so deleted or modified files can be restored.
* use encrypted backups to protect sensitive data.

Achieving RTO & RPO

Recovery Time objective (RTO):- Reduce downtime by using standby servers & automated recovery tools to restore services quickly.

* Recovery Point Objective (RPO):- Minimize data loss through frequent backups & continuous data replication.

**Bussiness Continuity During Failures**

⇒ Monitor backup Status regulally

⇒ Test backup restoration periodically

⇒ Main disaster recovery documentation.

⇒ Train empolyees on disaster recovery procedure

**Conclusion:-** cloud backup, replication & versioning Protect data & reduce downtime. They ensure fast recovery & Continuous Service availability.

(4) **Government Agency - Hybrid cloud**

**Introduction:-** A government agency uses a hybrid cloud where sensitive data stays on premises while non sensitive applications are hosted in the public cloud.

**Hybrid cloud Architecture :-**

* Store Sensitive data & critical applications in the private cloud on premises data center

* Deploy non sensitive applications in the website & citizen services in the public cloud.

* Connect the private & public cloud using a secure VPN or dedicated network connection

* use cloud monitoring tools to monitor both environments from a single dashboard.

Identity Management & Encryption:-

To ensure Secure Communication:

* Implement Identity & Access management (IAM) to provide role based access Control.

* Enable Multifactor Authentication (MFA) for all users

* Encrypt data ate reset using cloud encryption services.

* Encrypt data in transit using SSL/TLS protocols.

Challanges in Hybrid cloud:-

* Managing both on premises & cloud environments.

* Ensuring regulartory compliance & data Privacy.

* Network latency b/w private & public cloud.

* Higher Mangement and maintainance complexity.

Advantages

* protects sensitive government data.
* Provides Scalability for non sensitive applications
* Reduces infrastructure costs by using public Cloud resources

Conclusion:- A hybrid cloud architecture provides Strong Security, regulatory Compliance & flexible resource management for agencies.

(5) Saas provider Multi-Region Depolyment

A Saas provider needs to ensure high availability & fault tolerance so that its application Continues to operate even if one Cloud region fails.

Multi Region Depbyment Strategy

the proposed cloud architecture includes

* Deploy the application in multi cloud regions.
* use a global load balancer to distribute user requests to nealest healthy region.

* Maintain identical application Servers in each region.

Data Replication & Failover

* Enable cross region data replication to keep data synchronized.
* Use backup storage to recover data in case of failures.
* Perform regular disaster recovery testing.

Monitoring & Alerting

* Configure automatic alerts for failures or unusal activity.
* Monitor application Performance the continuously.
* Generate logs & reports for troubleshooting & performance analysis.

Conclusion :-

A multi region cloud deployment with data replication & automatic failover ensure Continuous Service even during regional failures.